# FEVER: experimental results

**Table 1**
LFTO attack parameter setting.

| Attack | T1(s) | S(pkt/T) | M(pkt) | Pkt/refresh | Pkt size | T2(s) |
|---|---|---|---|---|---|---|
| A2 | [5.5,9] | 100 | 1700 | 1 | 64 B | 83.5 |
| A4 | [6,9.5] | 100 | 1700 | 1-3 | benign-like | 103.0 |
| E9P | [2,3] | 100 | 1700 | 1-3 | benign-like | 31.0 |

T2 is the average time from the start of the attack until the flow table overflows without defense.

**Table 2**
Attack detection results of FEVER.

| Attack | Attack detection rate | Attacker identification rate | Detection time(s) | False blocking rate |
|---|---|---|---|---|
| A2 | 100 % | 100 % | 70.7 | 0 % |
| A4 | 100 % | 100 % | 126.4 | 0 % |
| E9P | 100 % | 100 % | 70.0 | 0 % |
| No attack | - | - | - | 0 % |

**Table 3**
Results of malicious flow identification of LFTO attacks with different parameters.

| Attack | Accuracy | Recall | Precision | F1 Score |
|---|---|---|---|---|
| A2 | 99.39 % | 99.28 % | 99.95 % | 99.61 % |
| A4 | 21.18 % | 0.01 % | 14.29 % | 0.02 % |
| E9P | 10.98 % | 0.00 % | 0.00 % | 0.00 % |
| Average | 43.85 % | 33.10 % | 38.08 % | 33.21 % |
| Mix | 42.69 % | 30.66 % | 99.88 % | 46.92 % |

Per flow entry, FEVER's federated detector only. A4 and E9P are left to the volume block.

**Table 4**
Comparing the outcomes of various classifiers to recognize harmful flows.

| Classifier | Accuracy | Recall | Precision | F1 Score | F1 (seen remotely) | F1 (unseen) |
|---|---|---|---|---|---|---|
| FTOP | 96.94 % | 92.02 % | 99.79 % | 95.02 % | 81.16 % | 14.74 % |
| FTMaster | 97.07 % | 92.37 % | 99.79 % | 95.30 % | 82.20 % | 20.65 % |
| FloRa | 99.01 % | 97.50 % | 99.82 % | 98.62 % | 82.68 % | 28.46 % |
| DMS-BTT | 97.03 % | 92.27 % | 99.78 % | 95.22 % | 82.38 % | 19.94 % |

All classifiers at a 0.1 % false-positive rate on the same data. Seen remotely: attacks seen only by another controller; unseen: an attack no controller was trained on.

**Table 5**
Comparative experiment results.

| Method | Deploy plane | Average Pa | Max Pa | Average flow table usage | Earliest overflow time(s) | Packet-in message | Table-full message |
|---|---|---|---|---|---|---|---|
| None | None | 62.10 % | 94.43 % | 76.58 % | 21 | 101,854 | 3,217 |
| FTMaster | Control | 45.54 % | 82.16 % | 60.88 % | 31 | 111,955 | 1,509 |
| DMS-BTT | Control | 41.85 % | 78.24 % | 58.06 % | 40 | 105,310 | 271 |
| FEVER | Control | 21.94 % | 82.86 % | 40.32 % | 110 | 118,529 | 250 |

Mean per run over attacks A2, A4 and E9P; earliest overflow time is counted from the start of the attack (-: the table never overflowed).

**Table 5a**
Comparative experiment results under A2.

| Method | Deploy plane | Average Pa | Max Pa | Average flow table usage | Earliest overflow time(s) | Packet-in message | Table-full message |
|---|---|---|---|---|---|---|---|
| None | None | 58.43 % | 94.79 % | 75.20 % | 75 | 106,937 | 3,350 |
| FTMaster | Control | 9.98 % | 58.10 % | 30.21 % | - | 158,455 | 0 |
| DMS-BTT | Control | 18.67 % | 54.28 % | 38.55 % | - | 132,215 | 0 |
| FEVER | Control | 12.62 % | 84.74 % | 31.72 % | 110 | 113,602 | 622 |

**Table 5b**
Comparative experiment results under A4.

| Method | Deploy plane | Average Pa | Max Pa | Average flow table usage | Earliest overflow time(s) | Packet-in message | Table-full message |
|---|---|---|---|---|---|---|---|
| None | None | 54.77 % | 91.38 % | 70.94 % | 83 | 99,238 | 1,828 |
| FTMaster | Control | 53.36 % | 91.42 % | 69.06 % | 104 | 90,553 | 1,307 |
| DMS-BTT | Control | 47.58 % | 86.21 % | 64.63 % | 132 | 96,989 | 245 |
| FEVER | Control | 27.23 % | 80.04 % | 47.69 % | 208 | 118,623 | 103 |

**Table 5c**
Comparative experiment results under E9P.

| Method | Deploy plane | Average Pa | Max Pa | Average flow table usage | Earliest overflow time(s) | Packet-in message | Table-full message |
|---|---|---|---|---|---|---|---|
| None | None | 73.10 % | 97.13 % | 83.60 % | 21 | 99,386 | 4,472 |
| FTMaster | Control | 73.29 % | 96.97 % | 83.38 % | 31 | 86,856 | 3,219 |
| DMS-BTT | Control | 59.30 % | 94.23 % | 71.00 % | 40 | 86,725 | 568 |
| FEVER | Control | 25.96 % | 83.80 % | 41.56 % | 244 | 123,360 | 26 |

**Table 6**
Impact on benign users.

| Method | Benign flows harmed | Lost first packets | New connections succeeding | Decision time(ms) | Attackers blocked |
|---|---|---|---|---|---|
| None | 10.13 % | 28.2 % | 25.2 % | 30 | 0/30 |
| FTMaster | 4.80 % | 27.8 % | 23.6 % | 72 | 0/30 |
| DMS-BTT | 1.71 % | 32.6 % | 23.4 % | 65 | 0/30 |
| FEVER | 0.49 % | 19.8 % | 29.8 % | 13 | 30/30 |